

NICHOLAS A. EZZELL

QUANTUM INFORMATION THEORIST | HIGH-PERFORMANCE COMPUTING

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EDUCATION

- Ph.D. in Physics**, University of Southern California (Advisors: Itay Hen & Daniel Lidar) May 2025
U.S. Department of Energy Computational Science Graduate Fellow | GPA: 3.92 / 4.0
Dissertation: *Theory and design of algorithms for quantum systems*
- B.S. in Physics and Mathematics**, Mississippi State University May 2019
Goldwater Scholar | Presidential Scholar | Honors Thesis in Physics | GPA: 3.97 / 4.0

EXPERIENCE

- Visiting Scientist and Mentor, Information Sciences Institute of USC** April 2026–Present
- Co-designed a graduate research project in quantum spectral estimation, mentoring students through literature review, mathematical analysis, code implementation, and execution
- Quantum Information Scientist IV, HRL Laboratories** July 2025–April 2026
- Built 50× more efficient models of exchange-only qubits, enabling faster characterization of noisy dynamics through leakage-aware gate set tomography in collaboration with Sandia National Laboratory
 - Reduced computational errors by 100× by constructing noise-resilient dynamically decoupled logic gates
 - Contributed 20+ improvements to internal GitHub libraries, including bug fixes, Jupyter tutorials, and features that improved usability for 50+ users
- Doctoral Researcher, University of Southern California** August 2019–May 2025
- Identified optimal error-suppression strategies on superconducting qubits through systematic benchmarking, challenging theoretical expectations for dynamical decoupling
 - Made previously inaccessible quantum phase transitions detectable by extending quantum Monte Carlo to arbitrary observables using group-theoretic methods
- Graduate Fellow, Los Alamos National Laboratory** May 2021–August 2023
- Resolved a foundational question in quantum theory by formulating the low-rank approximation problem for quantum states and finding the optimal solution
 - Reduced quantum-state learning costs by 1500× in memory and 56× in samples with a variational alternative to state tomography tested on superconducting quantum computers
- Undergraduate Research Intern, Oak Ridge National Laboratory** May 2018–May 2019
- Built Python tools and custom annealing protocols for D-Wave quantum annealers, improving ground-state search performance and enabling reusable phase-diagram benchmarking workflows

SKILLS

Programming & Development: Python, Julia, C++, Bash, Git, Conda, Docker, CI/CD
Quantum Software: Qiskit, QuTiP, Stim, ITensor, PMR-QMC, Mathematica, HOQST.jl
Data and Machine Learning: Pandas, Clustering, Bootstrapping, PCA, Model Validation (AICc)
Scientific Computing: MPI, Slurm, Org-mode emacs, Jupyter, Matplotlib/Seaborn, L^AT_EX/Beamer

SELECTED PUBLICATIONS (see naezzell.github.io/publications for a full list)

- N. Ezzell and I. Hen (2026). “Advanced measurement techniques in quantum Monte Carlo: The permutation matrix representation approach,” *Computer Physics Communications*. GitHub: `advmeaPMRQMC`
- N. Ezzell, E. M. Ball, A. U. Siddiqui, M. M. Wilde, A. T. Sornborger, P. J. Coles, and Z. Holmes (2023). “Quantum mixed state compiling,” *Quantum Science and Technology*. GitHub: `qmsc`
- N. Ezzell, B. Pokharel, L. Tewala, G. Quiroz, and D. A. Lidar (2023). “Dynamical decoupling for superconducting qubits: a performance survey,” *Physical Review Applied*. GitHub: `edd`